

Institute of Actuaries of India

Subject SA5 – Finance

September 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) FCFF is enterprise level free cash flows available to both shareholders and bond holders. Free cash flow used here should be post tax operating free cash flows. The discount rate used here should be Weighted Average Cost of Capital (WACC). WACC is computed as follows

$$WACC = \frac{debt}{debt + equity} \times (1 - tax\%) \times R_{debt}^* + \frac{equity}{debt + equity} \times R_{equity}^*$$

where:

- $debt$ and $equity$ are measured in market value terms
- $tax\%$ is the corporation tax rate.
- $R_{equity}^* = R_f + \beta_{equity} (R_m - R_f)$
- $R_{debt}^* = R_f + \beta_{debt} (R_m - R_f)$

and $(R_m - R_f)$ represents the market risk premium.

FCFE is free cash flow to shareholders i.e. post tax operating cashflows after payment of interest obligation. Here the discount rate used should be the cost of equity (R_{equity}^*) which in turn is derived from CAPM.

Indicative Numerical example

Company with stable free cash flow to equity of 50 Cr (no growth) and debt of 200 Cr. Market risk premium of 6% and risk-free rate 7%. Tax rate of 30%.

FCFE	50.00	
Rf	7.00%	
Rm-Rf	6.00%	
Beta-equity	1.00	
Re	13.00%	
Beta - debt	-	
Rd	7.00%	
Debt	200.00	
Interest	14.00	
Tax Rate	30%	
MV Equity	384.62	FCFE/Re
EV using FCFE	584.62	MV Equity+Debt
FCFF	59.80	FCFE+interest*(1-tax rate)
WACC	10.23%	Formula for WACC
EV using FCFF	584.62	FCFF/WACC

- ii) In risky projects market value of debt can be volatile and materially different from book value of debt at different points of time making WACC computation difficult.

Government has given tax breaks to Reliable. So tax rate is also not stable during the course of the project. Project may generate losses initially gain tax shields and so on which makes estimation of tax rate used to calculate WACC difficult. FCFF method requires adjustment for tax shield on interest. FCFE can be modeled based on actual expected tax flows and makes it easier than FCFF method.

Equity inflow into projects can happen in multiple rounds in different years. Easier to model using FCFE approach. [5]

- iii) Common valuation approaches use a single discount rate across the project life. When debt to equity ratio changes every year with project cash flows used to repay debt the company becomes lesser risky.

Assuming debt beta to be close to zero as debt gets repaid

$$\beta_{equity} = \beta_{asset} \times \frac{D+E}{E} [2]$$

As leverage goes down equity beta starts trending lower towards the asset beta. As equity beta goes down cost of equity also goes down. Using the equity beta based on initial or average debt equity ratio of the project may hence lead to incorrect conclusions.

Since WACC uses cost of equity as an input the estimation of WACC can also be affected due to repayment of debt.

This can be corrected if multiple discount rates are used. Say the project is for 25 years then explicit cash flow projections are made for each of the 25 years along with expected debt repaid in each year. Based on the dynamically changing debt to equity ratio the discount rate is estimated each year and used. [9]

- iv) The WACC/Cost of equity depends on the leverage. This leverage needs to be computed on market value basis. According to efficient market hypothesis the NPV of the project needs to be considered as market value of equity rather than the equity that has been invested into the project. Since projects usually have positive NPV if this is not considered and only the equity that has been invested is considered then Market Value of equity is understated and hence leverage is overstated. This leads to higher estimate for cost of equity and WACC and makes the project look more unattractive than it actually is.

One problem here is the discount rate is needed to estimate NPV and NPV is needed to estimate the leverage to arrive at discount rate. This can be solved by using the fact that the project's equity value at the end of year 25 will be zero. An iterative procedure can be employed to find the NOV that when added to initial equity will lead to capital structure and discount rates that lead to equity value of zero at the end of 25 years. [7]

- v) Spread between yield on a bond issued by the country in a stable currency (say USD) and yield on a US government bond of similar maturity can be taken as a proxy for country specific credit risk. This can be added to the discount rate arrived at using CAPM

[3]

vi) Following are major covers expected

- Expropriation; when a host government limits or eliminates investor control over assets through confiscations and nationalizations.
- Capital controls; when an investor is unable to convert local currency into foreign exchange or transfer capital out of the country.
- Political violence; losses to business assets or activities due to war, terrorism, or civil disturbance.
- Breach of contract; when a host government fails to honor the terms of a contract or after an arbitration process awards the investor damages that the host government fails to pay.
- Failure of sovereign financial obligations; when a government fails to meet an unconditional financial obligation or guarantee.

[10]

vii) The following real options maybe considered to be there

The option to make follow-on investments if the immediate investment project succeeds

Reliable has the option to participate in future projects can have value in itself and may be added to the NPV of the project.

The option to abandon the project

Reliable may have the option to commit a smaller amount of capital make better assessment of profitability and risks involved before committing larger tranches of capital. It may have the option to abandon the project after the initial commitment which is very valuable. The flexibility involved is often valuable, as assets can be redirected elsewhere if events turn out to be worse than expected.

The option to vary the output or its production methods

The output of the project is oil which is a commodity with volatile pricing while costs are reasonably steady. Company may have the option to vary its output to maximize the returns depending on the price of the commodity.

[6]

viii) The crude oil price can be modeled using historical data say using a lognormal distribution.

Cost of production can be assumed to be reasonably stable with increase in line with some cost index

The debt service coverage ratio can be modeled for each of the 25 year estimated life of the project after super imposing the debt repayment schedule

A threshold for debt service coverage ratio, say 1.5X maybe used as warning trigger and it can be monitored as to when the trigger is crossed under stressed scenarios – say 95% confidence level

This will help the investors to schedule the equity cash in flows or decide on the dividend policy to be followed by the SPV. The mix between distributed cash and retained cash can be estimated more accurately

[5]

[60 Marks]**Solution 2:**

- i) The main forms of government policy that may affect the commercial and economic environment are:
1. Monetary policy – the control of some measures of the money supply and/or the level and structure of interest rates.
 2. Fiscal policy – decisions on the level and structure of taxation and government expenditure and hence the fiscal deficit.
 3. National debt management policy – the manipulation of the outstanding stock of government debt instruments held by the domestic private sector, in order to influence the level and structure of interest rates or the availability of liquid reserve assets to the banking sector.
 4. Exchange rate policy – directed towards achieving some target for the exchange rate of the domestic currency in terms of foreign currencies.
 5. Prices and incomes policy – aimed at influencing the rates of wage and price inflation.
 6. Taxation – decisions on the overall level of taxation and its distribution between personal direct, indirect, corporate and other (eg stamp duty).
 7. Competition policy – aimed at maintaining an appropriate operating environment for companies, eg by limiting monopoly powers.
 8. Labour policy – decisions that determine the flexibility of labour and the bargaining power of organised labour, eg minimum wage legislation.
 9. Government incentives for investment

Note: Students are expected to expand on above points to explain how these actions will help economy to bounce back to high growth.

[8]

- ii) A central bank may have control over:
1. monetary, interest rate, quantitative easing and inflation policy
 2. banking regulation
 3. implementation of government borrowing
 4. performance and integrity of financial markets, including lender of last resort
 5. intervention in currency markets
 6. printing and minting of notes and coins

It would be advisable to manipulate interest rates upwards if the Central Bank wishes to stop speculative funds from selling the domestic currency. This would help in protecting the currency exchange rate in respect of other global currencies.

One way to increase interest rates is via money market intervention. This is achieved through selling Treasury bills and government bonds to reduce the level of liquidity within the banking sector and hence increase short-term interest rates. Alternatively, it may increase bank base rates to directly control interest rates. Stopping quantitative easing would be another possible answer in the current environment.

Some of the other non-market (direct) controls that the Central Bank may also use such as:

1. setting minimum liquid reserve ratios for the domestic banks
2. setting interest rate ceilings for bank deposits
3. issuing directives regarding the types of lending to be undertaken. [8]

iii) Few Options available that I will suggest to Government of Valencia to manage the risk of large and untimely payouts/write-offs are as follows :

- a. Create a Disaster Fund and pre-fund/reserve in advance, i.e. transfer pre-defined amounts into a funding account and use this pot to pay out aid donations as needed
- b. Implement a matched pay as you go system where either donations are limited to any available budget surplus at the time
- c. Issue more conventional government bonds to provide the funding as and when needed
- d. Have a policy to limit (or stop) aid donations
- e. Restrict it to certain types of events, etc
- f. Stagger aid donations over a period of time
- g. Buy approximate hedges for disasters in other countries generally
- h. Transfer the risk through weather derivatives / loss warrants / catastrophe derivatives
- i. Set up an SPV with full capital structure from equity layer to senior debt layers
- j. Sell traditional catastrophe bonds
- k. Use a captive insurance operation
- l. Contract with a reinsurance company
- m. 'Co-insure' with other parties – e.g. other governments or aid agencies
- n. Encourage relevant private charitable donations through e.g. tax incentives
- o. Improve disaster forecasting and disaster management [8]

iv) The factors that need to be taken into consideration for the design and issuance of the Parametric Weather Bonds are as follows :

General Consideration:

- What currencies to offer the bond in – local or foreign
- Whether any collateral will be offered (unlikely but theoretically possible)
- The appropriate duration / term for the bond
- Whether the coupon is fixed or floating
- Target market – retail vs institutional
- Any need for roadshows and marketing
- How it will be sold (e.g. by auction / through brokers)
- Credit rating / working with rating agencies to rate this issue
- Tax status
- Any optionality in the bond (e.g. whether callable or not)
- Cost of issue

- Whether there will be a series of issues
- Market appetite for this type of security
- Structuring of the bond (SPV or other structure)
- Regulatory issues around the listing itself and afterwards eligibility for banks, insurance companies etc.
- Size of issue (also in relation to existing government debt market)

Specific Consideration:

- How to control the government's ability to anti-select against bondholders by subsequently having an overly generous aid programme, including:
 - Definition of aid payments and possible limitations by region or type of event
 - Any restrictions/guarantees given in relation to the level of aid payments that will be made
 - Guaranteed minimum maturity payment of zero / clarity on what happens if cumulative aid payments exceed defined principal
- Balance between what would be advantageous for the government vs what would make this more appealing to the end investor;
- How to model the expected size and frequency of aid payments
- ... including information on historic aid donation levels by region and events
- Selecting the initial yield to maturity (coupon rate plus any issue discount) to compensate for the expected "haircut" at maturity, any inherent sovereign default risk and for the complexity of the bond (compared to alternatives)
- Covenants are unlikely but may exist in relation to other aid programmes
- Potential difficulty of valuation, especially while a disaster is ongoing
- Formula for accumulation of aid pay-outs (and hence deduction from redemption proceeds)

[8]

v) Some of the risk and risk and attractiveness of such a bond to institutional investors are as follows :

- Yield offered by may be higher than other Bonds
- They are issued by Government of Valencia and hence will be highly secured.
- Institutional investors investing in Parametric Weather bonds are making excellent profits. Low volatility for a risky asset is one of the reasons. Some hedge funds choose places with low probability of having a catastrophe event. This guarantees yield.
- Parametric Weather Bonds provide great opportunities for investment portfolios. Parametric Weather Bonds will be uncorrelated to their other assets in the portfolio and help in diversification. These bonds have virtually zero correlation with conventional bonds, equities or the ups and downs of the market. Cat bonds have been able to add diversification to investors' portfolios and make new ways of generating money.
- Multi-year commitments-capacity and pricing
Traditional reinsurance is offered for one-year coverage basis. Ceding insurer must re negotiate contract each year. There may be capacity and pricing uncertainty each year. Whereas cat bonds are issued for a two or three year maturity period, providing fully collateralized coverage at a fixed price for that period.
- Reinsurance trigger familiarity

It is difficult to under the payout in the case of event and may cause basis risk for investors.

- Collection risk avoidance

The bonds are fully collateralized and the bonds protected by a reinsurance trust for the benefit of the ceding insurer. Considering documentation is drafted properly and facility administered, the insurers face minimal risk in collecting reinsurance claims in cat bonds.

- Lower long term transactional costs

Due to long duration of cat bonds, insurers can spread out transactional costs of putting a risk transfer mechanism,

- Increased market capacity:

- i. Cat bonds provides opportunity for occasional participants in the risk transfer market provide additional capacity based on their own business plan, without any additional costs which may obtain had they participated in traditional reinsurance facilities

[8]

[40 Marks]
